MLOPS Assignment-2

Tracking a Pre-trained Sentiment Model with MLflow

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1. Which run had the lowest latency, and why might that be?

In the first execution, run_batch4 had the lowest latency of 0.034 seconds, followed by run_batch8 (0.047 seconds) and run_batch1 (0.163 seconds).

Processing multiple sentences together reduces repeated inference-call overhead. The exact timings may vary between executions due to model warm-up and system conditions.

2. Did accuracy change between runs? Why or why not?

No. Accuracy remained 1.0 (100%) for all three batch sizes.

The same pretrained sentiment-analysis model and test sentences were used in all runs. Changing the batch size only affects how inputs are grouped for inference; it does not change the model weights or expected predictions.

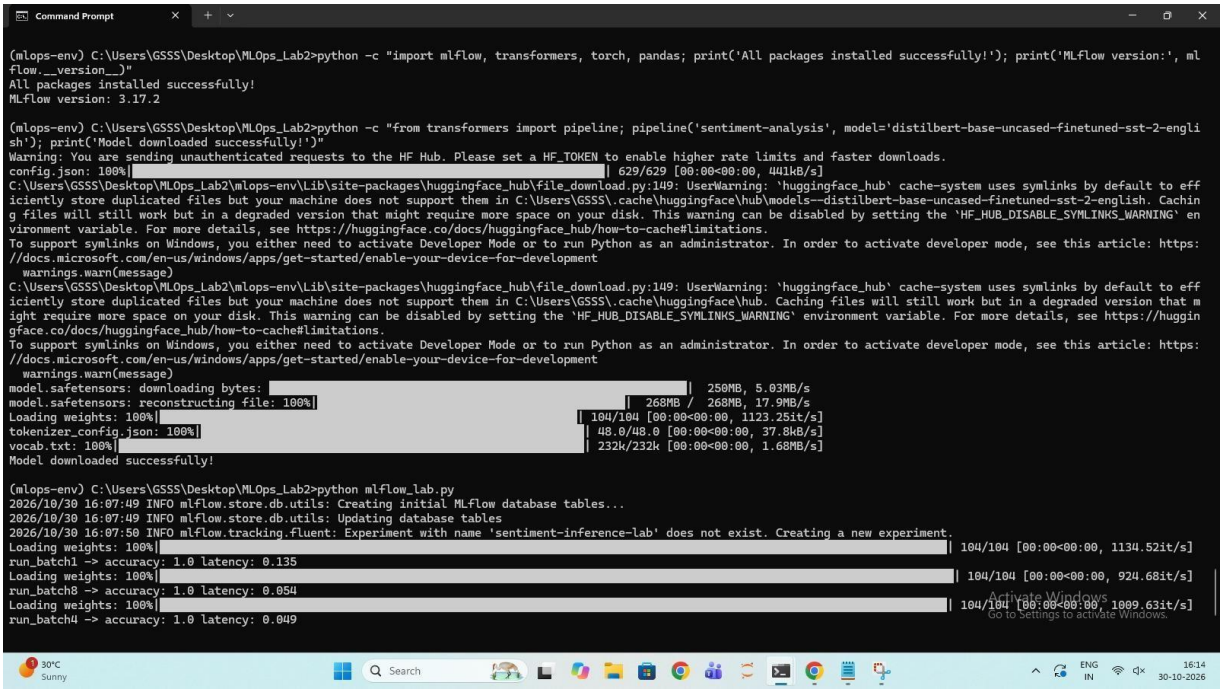
3. Name two benefits of using MLflow here instead of printing results in the terminal.

1. Reproducibility and Record-Keeping:
MLflow saves parameters, metrics, and artifacts such as predictions.csv for future reference and inspection.

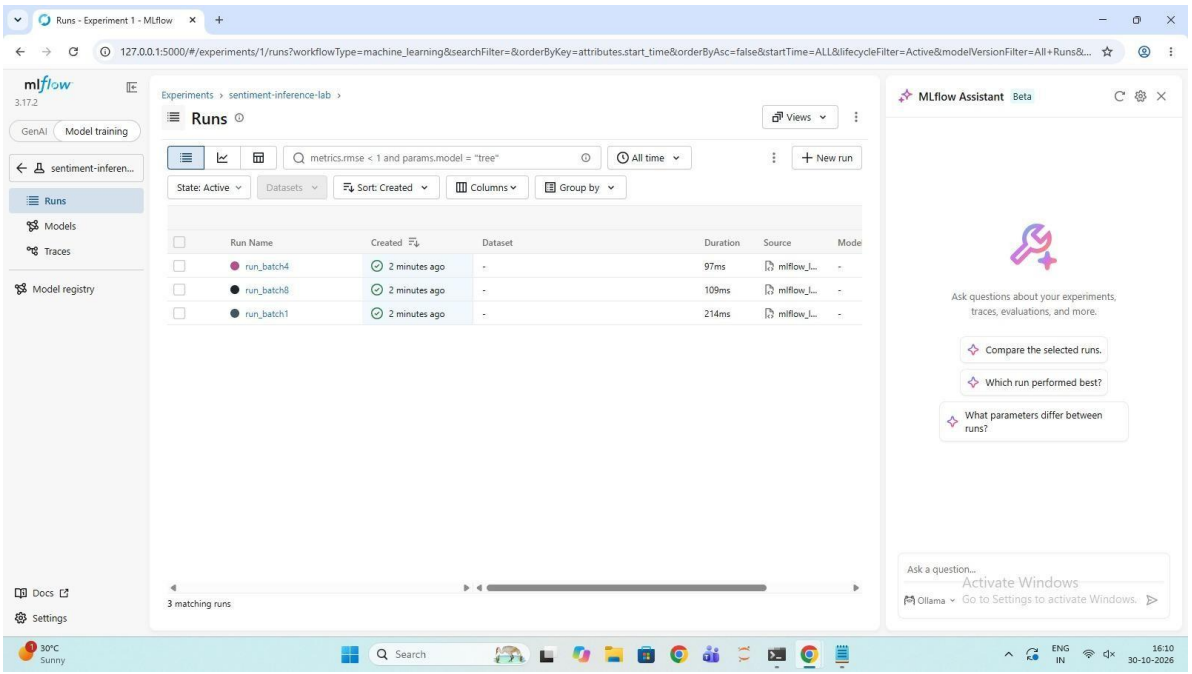
2. Easy Comparison:
MLflow displays multiple runs and their metrics side by side, making it easier to compare accuracy, confidence, and latency than reading terminal output alone.

TOOLS AND TECHNOLOGIES USED:

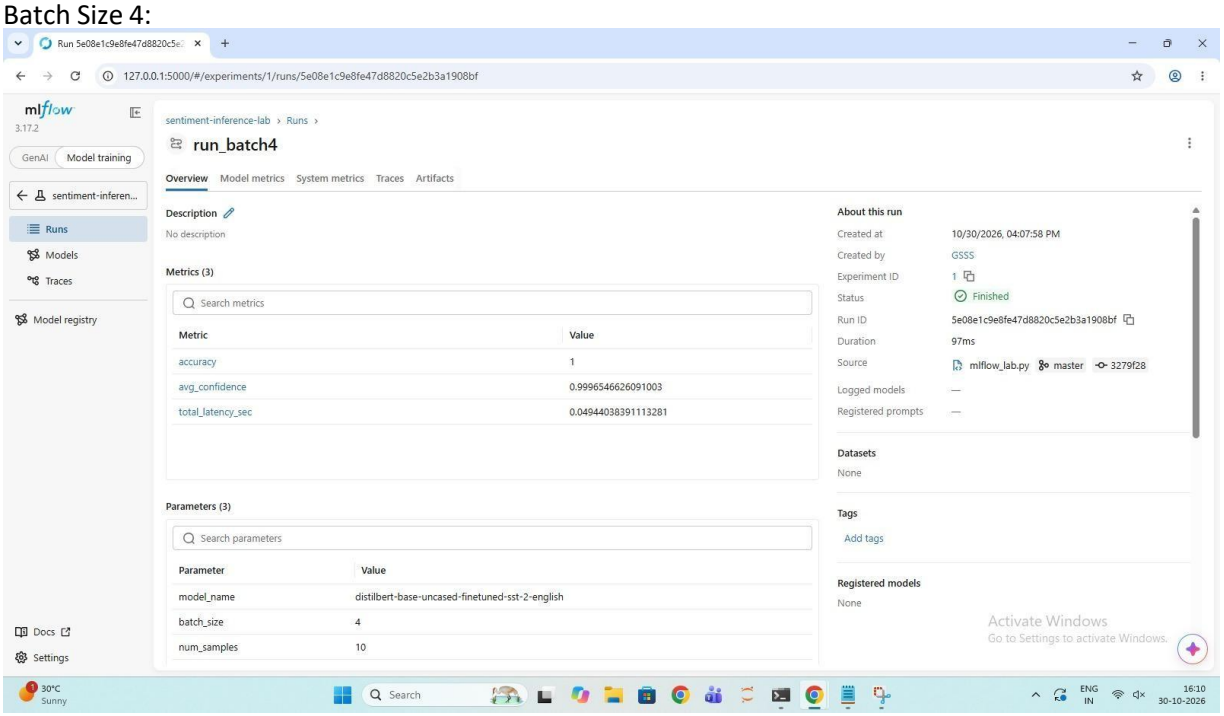
- Windows 11
- Python 3.12
- Visual Studio Code
- MLflow 3.17.2
- Pandas
- DistilBERT Pre-trained Model
- Python Virtual Environment (venv)
- Command Prompt (CMD)
- MLflow Web UI (localhost:5000)


Command Prompt
(mlops-env) C:\Users\GSSS\Desktop\MLOps_Lab2>python -c "import mlflow, transformers, torch, pandas; print('All packages installed successfully!'); print('MLflow version:', mlflow.__version__)"
All packages installed successfully!
MLflow version: 3.17.2
(mlops-env) C:\Users\GSSS\Desktop\MLOps_Lab2>python -c "from transformers import pipeline; pipeline('sentiment-analysis', model='distilbert-base-uncased-finetuned-sst-2-english'); print('Model downloaded successfully!')"
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
config.json: 100%| 629/629 [00:00<00:00, 441kB/s]
C:\Users\GSSS\Desktop\MLOps_Lab2\mlops-env\Lib\site-packages\huggingface_hub\file_download.py:149: UserWarning: `huggingface_hub` cache-system uses symlinks by default to efficiently store duplicated files but your machine does not support them in C:\Users\GSSS\.cache\huggingface\hub\models--distilbert-base-uncased-finetuned-sst-2-english. Caching files will still work but in a degraded version that might require more space on your disk. This warning can be disabled by setting the `HF_HUB_DISABLE_SYMLINKS_WARNING` environment variable. For more details, see https://huggingface.co/docs/huggingface_hub/how-to-cache#limitations.
To support symlinks on Windows, you either need to activate Developer Mode or to run Python as an administrator. In order to activate developer mode, see this article: https://docs.microsoft.com/en-us/windows/apps/get-started/enable-your-device-for-development
warnings.warn(message)
C:\Users\GSSS\Desktop\MLOps_Lab2\mlops-env\Lib\site-packages\huggingface_hub\file_download.py:149: UserWarning: `huggingface_hub` cache-system uses symlinks by default to efficiently store duplicated files but your machine does not support them in C:\Users\GSSS\.cache\huggingface\hub. Caching files will still work but in a degraded version that might require more space on your disk. This warning can be disabled by setting the `HF_HUB_DISABLE_SYMLINKS_WARNING` environment variable. For more details, see https://huggingface.co/docs/huggingface_hub/how-to-cache#limitations.
To support symlinks on Windows, you either need to activate Developer Mode or to run Python as an administrator. In order to activate developer mode, see this article: https://docs.microsoft.com/en-us/windows/apps/get-started/enable-your-device-for-development
warnings.warn(message)
model.safetensors: downloading bytes: 250MB, 5.03MB/s
model.safetensors: reconstructing file: 100%| 268MB / 268MB, 17.9MB/s
Loading weights: 100%| 104/104 [00:00<00:00, 1123.25it/s]
tokenizer_config.json: 100%| 48.0/48.0 [00:00<00:00, 37.8kB/s]
vocab.txt: 100%| 232k/232k [00:00<00:00, 1.68MB/s]
Model downloaded successfully!
(mlops-env) C:\Users\GSSS\Desktop\MLOps_Lab2>python mlflow_lab.py
2026/10/30 16:07:49 INFO mlflow.store.db.utils: Creating initial MLflow database tables...
2026/10/30 16:07:49 INFO mlflow.store.db.utils: Updating database tables
2026/10/30 16:07:50 INFO mlflow.tracking.fluent: Experiment with name 'sentiment-inference-lab' does not exist. Creating a new experiment.
Loading weights: 100%| 104/104 [00:00<00:00, 1134.52it/s]
run_batch1 -> accuracy: 1.0 latency: 0.135
Loading weights: 100%| 104/104 [00:00<00:00, 924.68it/s]
run_batch8 -> accuracy: 1.0 latency: 0.054
Loading weights: 100%| 104/104 [00:00<00:00, 1009.63it/s]
run_batch4 -> accuracy: 1.0 latency: 0.049
Activate Windows
Go to Settings to activate Windows.
30°C
Sunny
Search
ENG
IN
16:14
30-10-2026

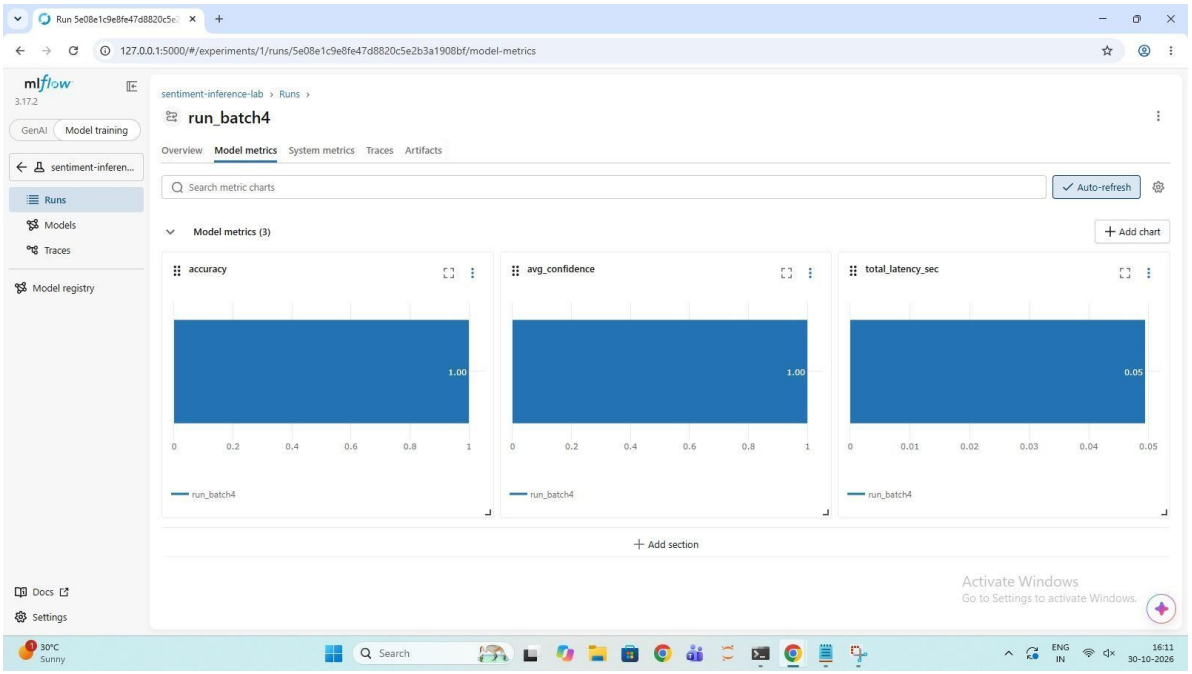
Dashboard


Runs - Experiment 1 - MLflow
127.0.0.1:5000/#/experiments/1/runs?workflowType=machine_learning&searchFilter=&orderByKey=attributes.start_time&orderByAsc=false&startTime=ALL&lifecycleFilter=Active&modelVersionFilter=All+Runs&...
mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
Docs
Settings
Experiments › sentiment-inference-lab ›
Runs
Views
metrics.rmse < 1 and params.model = "tree"
All time
New run
State: Active
Datasets
Sort: Created
Columns
Group by
Run Name
Created
Dataset
Duration
Source
Mode
run_batch4
2 minutes ago
97ms
run_batch8
2 minutes ago
109ms
run_batch1
2 minutes ago
214ms
3 matching runs
MLflow Assistant
Beta
Ask questions about your experiments, traces, evaluations, and more.
Compare the selected runs.
Which run performed best?
What parameters differ between runs?
Ask a question...
Ollama
Activate Windows
Go to Settings to activate Windows.
30°C
Sunny
Search
ENG
IN
16:10
30-10-2026

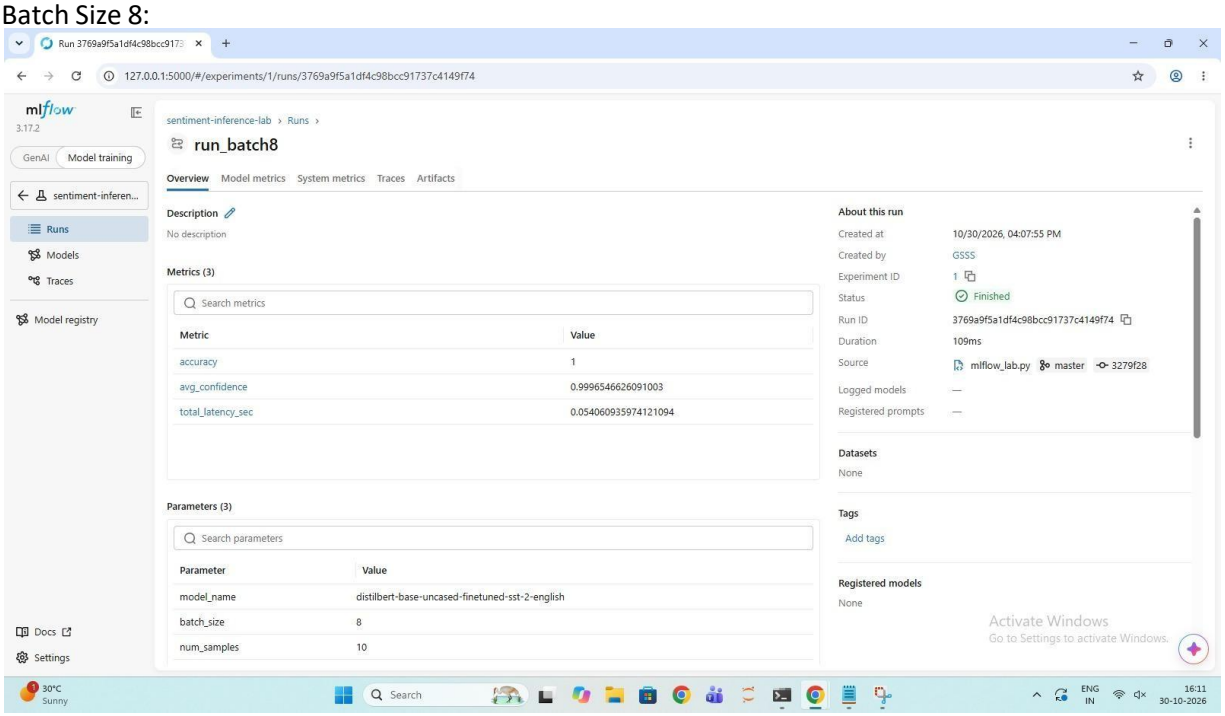
Batch Size 4:



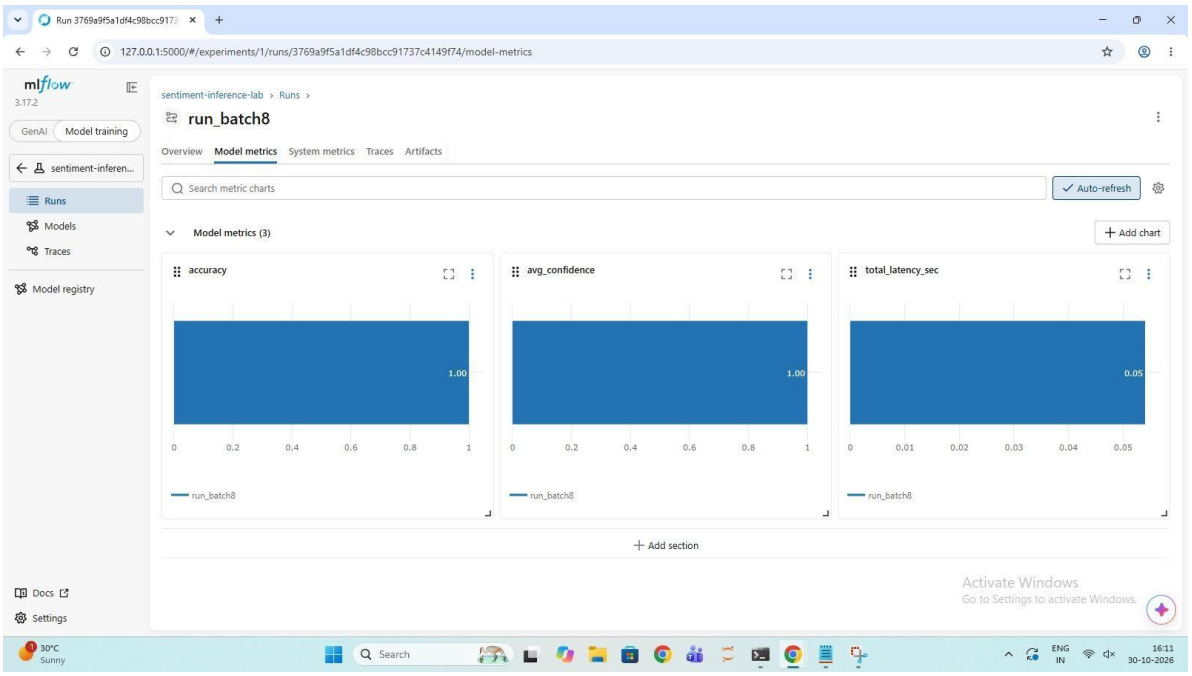
Performance Metrics and Parameters of Batch Size 4:



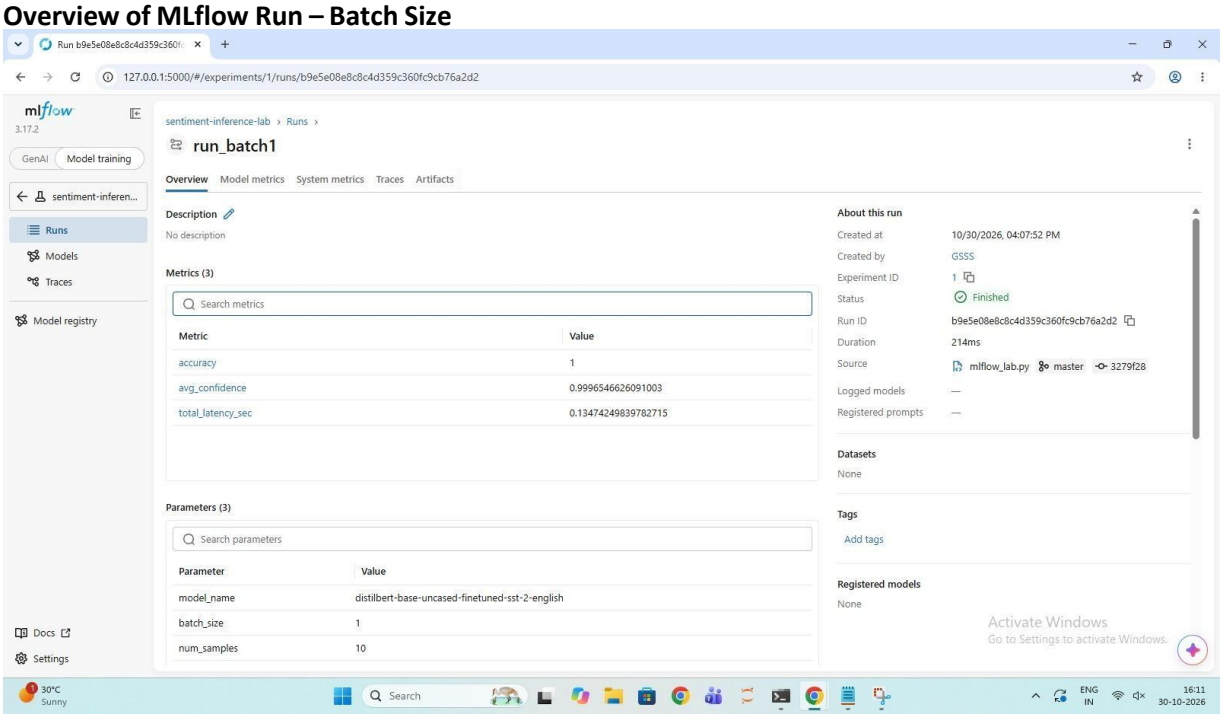
Batch Size 8:



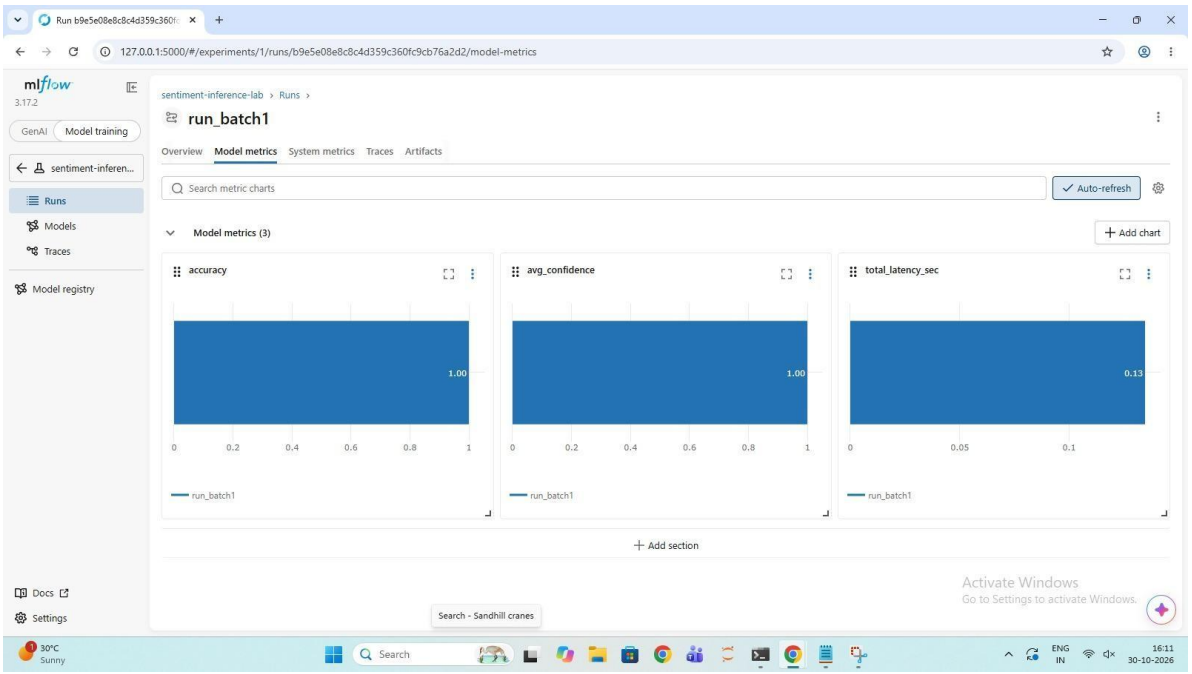
Model Performance Metrics for Batch Size 8:



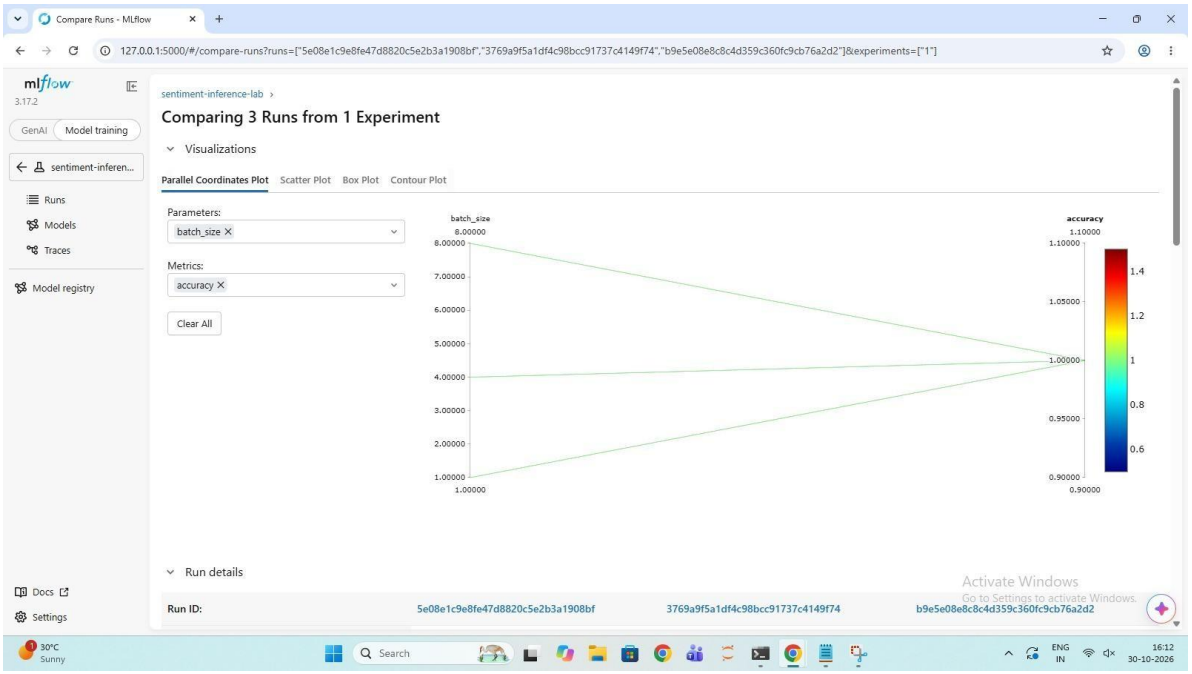
Overview of MLflow Run – Batch Size

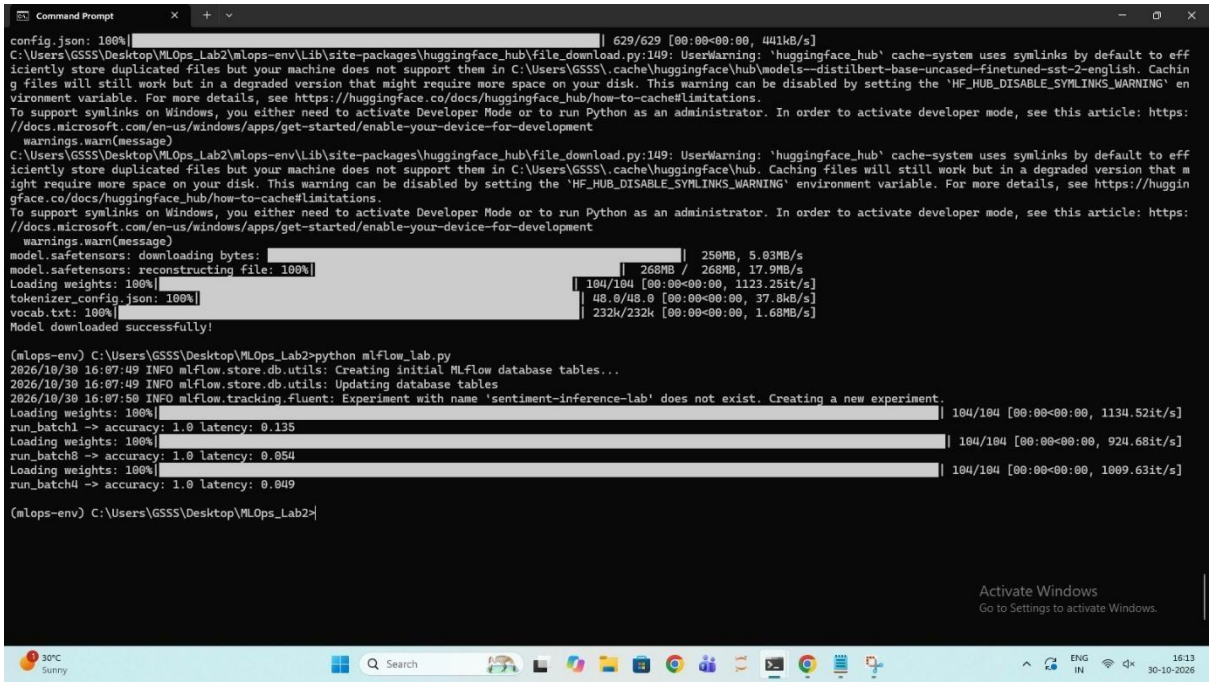


Model Performance Metrics for Batch Size 1:



Comparison of Three MLflow Experiment





MLflow Parameter and Metric Logging in Python